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TRANSPORT PHENOMENA II, CHE 312

EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A Newtonian fluid with constant physical properties is in a laminar flow in a circular tube of inner radius R according to the following velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

The fluid is highly viscous, leading to substantial energy dissipation due to viscous effects.

Answer the following questions in terms of the constant parameters:

$v_{z,max}$: maximum velocity of the fluid

R : tube's inner radius

μ : viscosity of the liquid

ρ : density of the liquid

c_p : isobaric heat capacity per unit mass of the fluid

k : thermal conductivity of the fluid

- 20/20 1- Using the **appropriate** equation of energy and the dissipation function, write a partial differential equation (PDE) describing the fluid's temperature profile as a function of r and z .

If the temperature of the tube's inner surface is maintained constant at T_0 , the asymptotic (far enough from the inlet) temperature profile within the fluid becomes independent of z .

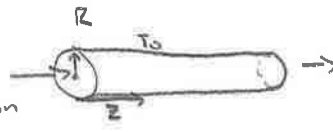
- 10/10 2- Write an ordinary differential equation (ODE) describing the fluid's asymptotic temperature profile as a function of r .

- 30/30 3- Solve the above ODE and apply appropriate boundary conditions to find the fluid's asymptotic temperature profile as a function of r .

- 5/20 4- Determine the fluid's bulk temperature T_b based on the asymptotic temperature profile derived above.

- 5/20 5- Calculate the Nusselt number $Nu = \frac{2hR}{k}$ using the heat transfer coefficient based on the asymptotic bulk temperature $q_r|_{r=R} = h(T_b - T_0)$.

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$$1) \quad v_z = v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

$$\rho C_p \left(v_z \frac{\partial T}{\partial z} \right) = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \cancel{\frac{\partial^2 T}{\partial z^2}} + \mu \psi_v$$

$$\psi_v = \left(\frac{\partial v_z}{\partial r} \right)^2 \rightarrow \frac{\partial v_z}{\partial r} = \frac{\partial}{\partial r} \left(v_{z,max} - v_{z,max} \left(\frac{r^2}{R^2} \right) \right) \rightarrow \frac{\partial v_z}{\partial r} = -2 v_{z,max} \frac{r}{R^2}$$

$$\hookrightarrow \psi_v = \left(\frac{\partial v_z}{\partial r} \right)^2 = \frac{4}{R^2} v_{z,max}^2 \left(\frac{r^2}{R^2} \right)$$

$$\hookrightarrow v_{z,max} \left[1 - \left(\frac{r^2}{R^2} \right) \right] \rho C_p \frac{\partial T}{\partial z} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{4\mu}{R^2} v_{z,max}^2 \left(\frac{r^2}{R^2} \right)$$

$$2) \quad \frac{\partial T}{\partial z} \rightarrow \text{large } z=0 \rightarrow 0 = \frac{k}{r} \cdot \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{4\mu}{R^2} v_{z,max}^2 \left(\frac{r}{R} \right)^2$$

$$3) \quad -\frac{k}{r} \cdot \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = \frac{4\mu}{R^2} v_{z,max}^2 \left(\frac{r}{R} \right)^2$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = -\frac{4\mu}{R^2 k} v_{z,max}^2 \frac{r^3}{R^2} \rightarrow r \frac{\partial T}{\partial r} = -\frac{4\mu}{R^2 k} v_{z,max}^2 \cdot \frac{1}{4} \frac{r^4}{R^2} + C_1$$

$$\hookrightarrow \frac{\partial T}{\partial r} = -\frac{\mu}{R^2 k} v_{z,max}^2 \frac{r^3}{R^2} + \frac{C_1}{r}$$

$$\frac{\partial T}{\partial r} (r=0) = 0 \rightarrow C_1 = 0$$

$$\hookrightarrow \frac{\partial T}{\partial r} = -\frac{\mu}{R^2 k} v_{z,max}^2 \frac{r^3}{R^2} \rightarrow T(r) = -\frac{\mu}{4k} v_{z,max}^2 \left(\frac{r}{R} \right)^4 + C_2$$

$$T(r=R) = T_0 \rightarrow T_0 = -\frac{\mu}{4k} v_{z,max}^2 + C_2 \rightarrow C_2 = T_0 + \frac{\mu}{4k} v_{z,max}^2$$

$$\hookrightarrow T(r) - T_0 = \frac{\mu}{4k} v_{z,max}^2 \left[1 - \left(\frac{r}{R} \right)^4 \right]$$

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4)

$$T(r) - T_0 = \frac{\mu}{4k} N_z^2 \quad bc \quad N_z^2 = N_{z,max}^2 \left[1 - \left(\frac{r}{R}\right)^4\right]$$

$$\hookrightarrow T_b - T_0 = \frac{\mu}{4k} \bar{N}_z^2 \quad \Rightarrow \bar{N}_z = N_{z,avg}$$

$$\hookrightarrow T_b = T_0 + \frac{\mu}{4k} \bar{N}_z^2$$

$$q_r = -k \frac{\partial T}{\partial r}$$

5)

$$q_r|_{r=R} = h(T_b - T_0)$$

$$\frac{k \mu}{R^2 k} N_{z,max}^2 \frac{r^3}{R^2} \Big|_{r=R} = h(T_b - T_0)$$

$$\frac{\mu}{R} N_{z,max}^2 = h(T_b - T_0)$$

$$T_b - T_0 = T_0 - T_0 + \frac{\mu}{4k} \bar{N}_z^2$$

$$\frac{\mu}{Rh} N_{z,max}^2 = T_b - T_0$$

$$\hookrightarrow \frac{\mu}{Rh} N_{z,max}^2 = \frac{\mu}{4k} \bar{N}_z^2$$

$$\frac{4k}{Rh} = \frac{\bar{N}_z^2}{N_{z,max}^2} \rightarrow \frac{2Rh}{8k} = \frac{N_{z,max}^2}{\bar{N}_z^2} \rightarrow$$

$$\boxed{\frac{1}{8} Nu = \frac{N_{z,max}^2}{\bar{N}_z^2}}$$